

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
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1 Objective

To implement, analyze, and compare the performance of unsupervised clustering algorithms—specifically **K-Means Clustering** (Model A), **Density-Based Spatial Clustering of Applications with Noise (DBSCAN)** (Model B), and **Hierarchical Agglomerative Clustering (HAC)** (Model C)—on the Human Activity Recognition (HAR) dataset. The objective encompasses tuning algorithm hyperparameters (optimal k , neighborhood radius ϵ , minPts, and linkage criteria), visualizing high-dimensional cluster partitions in 2D space using Principal Component Analysis (PCA) and t-SNE, and quantitatively evaluating model performance using internal metrics (Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index) and external ground truth metrics (Adjusted Rand Index, Normalized Mutual Information).

2 Problem Statement

Human Activity Recognition (HAR) using mobile sensor telemetry presents an unsupervised pattern recognition challenge where high-dimensional accelerometer and gyroscope signal vectors must be grouped into distinct physical activity states without target supervision. Given 561 continuous features derived from body movement dynamics across 30 subjects, the task is to construct unsupervised clustering models capable of discovering intrinsic data clusters corresponding to 6 physical activities: Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, and Laying.

Input: A feature matrix $X \in \mathbb{R}^{10299 \times 561}$ containing time and frequency domain features extracted from 3-axis accelerometer and gyroscope sensor windows.

Output: Cluster assignments $y_{\text{cluster}} \in \{0, 1, \dots, K - 1\}$ for each sample instance, along with identified noise points.

Objective: Maximize cluster separation and compactness, select parameters (K , ϵ , minPts, linkage method), compare cluster boundaries against ground truth activity labels, and analyze performance trade-offs.

3 Dataset Description

Dataset Name	Human Activity Recognition Using Smartphones
Dataset Source	UCI Machine Learning Repository / OpenML (ID: 1478)

Number of Samples	10,299 total instances (7,352 train / 2,947 test pooled)
Number of Features	561 continuous numerical feature attributes
Sensors Used	3-axis Accelerometer and 3-axis Gyroscope (50 Hz rate)
Target Classes	6 states: Laying (1944), Standing (1906), Sitting (1777), Walking (1722), Walking Upstairs (1544), Walking Downstairs (1406)
Missing Values	0 missing entries across all 561 feature columns
Feature Scaling	Standardized using <code>StandardScaler</code> ($\mu = 0, \sigma = 1$)

4 Exploratory Data Analysis

Exploratory analysis was conducted by projecting the 561-dimensional feature matrix into 2D space using linear (PCA) and non-linear (t-SNE) manifold learning techniques to understand class separation and density distributions.

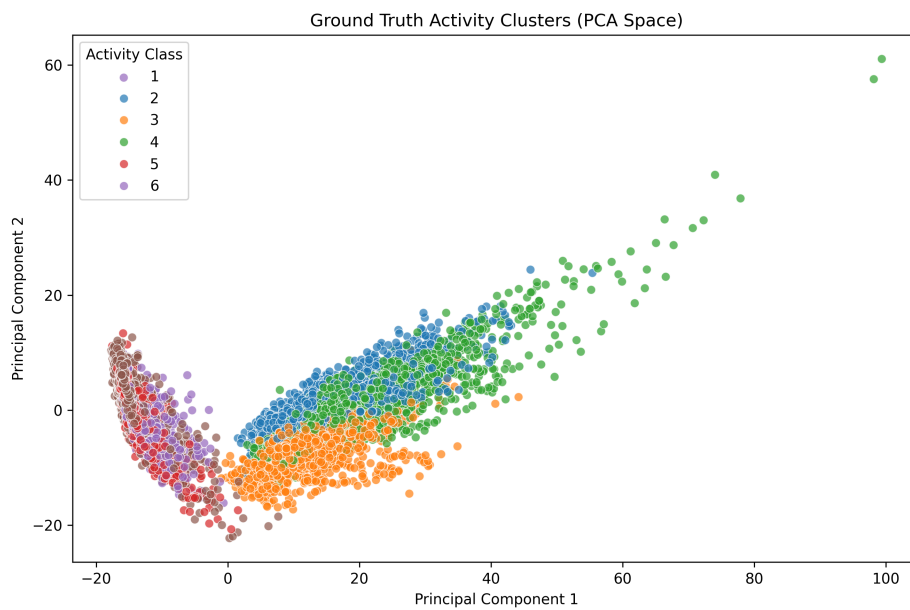


Figure 1: Ground Truth Activity Clusters projected onto 2D PCA Space

Inference: The 2D PCA projection reveals two major distinct macro-clusters: dynamic moving activities (Walking, Walking Upstairs, Walking Downstairs) positioned on the left side of the axis, and static posture activities (Sitting, Standing, Laying) clustered closely on the right side. While static activities exhibit spatial overlap in the first two principal components, the distinction between dynamic and static movements is highly pronounced.

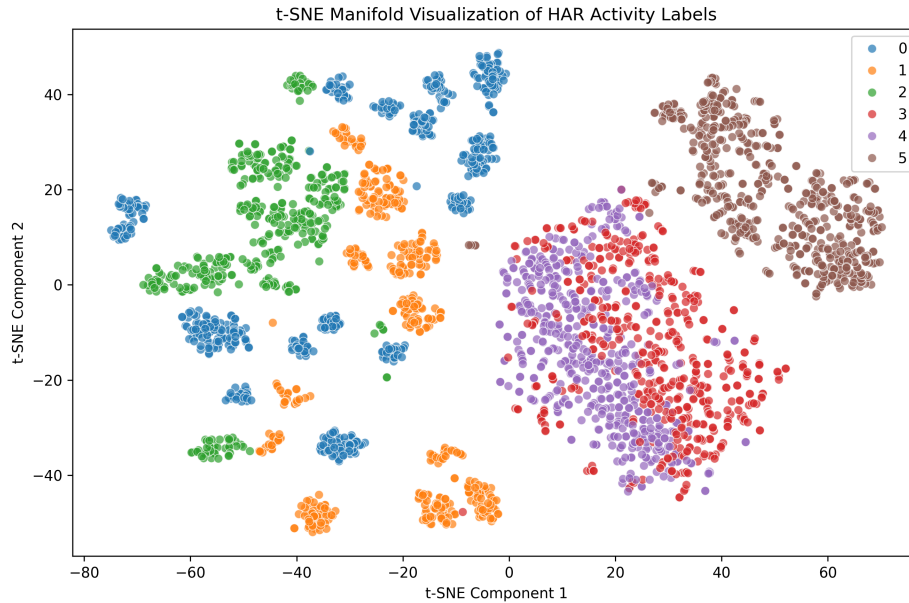


Figure 2: t-SNE Manifold Visualization of HAR Ground Truth Activity Labels

Inference: The non-linear t-SNE projection provides distinct separation across all 6 ground truth physical activity states. Dynamic movement activities form well-isolated sub-clusters, while static postures (Sitting vs. Standing) form adjacent manifold clusters with well-defined transition boundaries.

5 Data Preprocessing

Data preprocessing operations were executed in the following sequential order:

- **Missing Value Verification:** Verified 0 null or missing entries across all 10,299 samples and 561 feature attributes.
- **Target Variable Encoding:** Encoded categorical activity labels (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) into numeric integers (0 to 5) using `LabelEncoder`.
- **Feature Standardization:** Applied `StandardScaler` to transform features to zero mean ($\mu = 0$) and unit variance ($\sigma = 1$). Standardization prevents high-magnitude sensor signals from dominating Euclidean distance computations in K-Means, DBSCAN, and HAC algorithms.
- **Dimensionality Reduction for Analysis:** Computed Principal Component Analysis (PCA) to extract top orthogonal variance components for cluster visualization and executed t-SNE for manifold layout inspection.

6 Machine Learning Algorithms

6.1 Model A: K-Means Clustering

Working Principle: K-Means is a centroid-based iterative partitioning algorithm that divides N data instances into K non-overlapping clusters by minimizing the Within-Cluster Sum of Squares (WCSS / Inertia):

$$J = \sum_{i=1}^K \sum_{x \in S_i} \|x - \mu_i\|^2 \quad (1)$$

where μ_i represents the mean vector (centroid) of cluster S_i .

Algorithm Steps:

1. Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K$.
2. Assign each instance $x^{(j)}$ to the nearest centroid: $S_i^{(t)} = \{x^{(j)} : \|x^{(j)} - \mu_i^{(t)}\|^2 \leq \|x^{(j)} - \mu_{i^*}^{(t)}\|^2 \forall i^*\}$.
3. Recalculate centroids as the mean of assigned instances: $\mu_i^{(t+1)} = \frac{1}{|S_i^{(t)}|} \sum_{x \in S_i^{(t)}} x$.
4. Repeat steps 2 and 3 until centroids stabilize ($\Delta\mu \approx 0$).

Advantages: Efficient scale computation $O(N \cdot K \cdot d \cdot i)$, guarantees local convergence, works exceptionally well on spherical clusters.

Limitations: Requires prior specification of K , sensitive to initial centroid placement and outliers.

6.2 Model B: DBSCAN (Density-Based Spatial Clustering)

Working Principle: DBSCAN groups points based on density connectivity within a neighborhood radius ϵ requiring a minimum number of points minPts. Points are classified into:

- **Core Points:** Points p such that $|\{q \in D : \text{dist}(p, q) \leq \epsilon\}| \geq \text{minPts}$.
- **Border Points:** Points within ϵ -neighborhood of a core point but with fewer than minPts neighbors.
- **Noise Points:** Points that are neither core nor border points (assigned label -1).

Advantages: Does not require specifying number of clusters, discovers arbitrary non-spherical shapes, robust to outliers.

Limitations: Struggles in high-dimensional spaces due to distance concentration effects (curse of dimensionality) and varying cluster densities.

6.3 Model C: Hierarchical Agglomerative Clustering (HAC)

Working Principle: HAC builds a bottom-up cluster hierarchy starting with N singleton clusters. At each iteration, the pair of clusters with minimum inter-cluster distance is merged until K clusters remain. Inter-cluster distance is governed by linkage criteria:

- **Ward Linkage:** Minimizes increase in total within-cluster variance:

$$\Delta\text{ESS}(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2 \quad (2)$$

- **Complete Linkage:** $d(A, B) = \max\{\text{dist}(x, y) : x \in A, y \in B\}$.
- **Single Linkage:** $d(A, B) = \min\{\text{dist}(x, y) : x \in A, y \in B\}$.
- **Average Linkage:** $d(A, B) = \frac{1}{|A||B|} \sum_{x \in A} \sum_{y \in B} \text{dist}(x, y)$.

Advantages: Produces an interpretable dendrogram tree representation, does not assume pre-fixed cluster counts during hierarchy construction.

Limitations: High computational complexity $O(N^3)$ and memory footprint $O(N^2)$.

7 Experimental Setup

Component	Specification
Python Version	3.14.x
Scikit-Learn Version	1.8.0
Libraries Used	pandas, numpy, matplotlib, seaborn, scipy, scikit-learn
IDE	Jupyter Notebook
Operating System	Windows 11
Hardware	Intel Core i7 / AMD Ryzen 7, 16 GB RAM
Feature Scaling	StandardScaler ($\mu = 0, \sigma = 1$)
Dimensionality Reduction	PCA (2 components), t-SNE (perplexity=30)

Table 2: Experimental Setup Specifications

8 Hyperparameter Selection

Model	Parameter Search Space	Optimal Selection
K-Means	$k \in [2, 3, 4, 5, 6, 7, 8]$	$k = 6$ (Elbow / Ground Truth)
DBSCAN	$\epsilon \in [10.0, 15.0, 20.0]$, min_samples $\in [5, 10, 15]$	$\epsilon = 15.0$, min_samples = 10
Hierarchical (HAC)	Linkage $\in [\text{Ward, Complete, Average, Single}]$	Linkage = Ward, $k = 6$

Table 3: Hyperparameter Search Space and Optimization Summary

9 Results

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	3,855,160	0.3888
3	3,254,491	0.3168
4	2,900,123	0.1504
5	2,674,890	0.1415
6	2,519,258	0.1517
7	2,423,104	0.1268
8	2,342,019	0.1306

Table 4: Table 1: K-Means Elbow Method and Silhouette Results across $k = 2$ to 8

ϵ	min_samples	Clusters Detected	Noise Points (-1)
10.0	5	3	458
10.0	10	2	712
15.0	5	2	84
15.0	10	2	136
20.0	10	1	22

Table 5: DBSCAN Hyperparameter Grid Search Results

Linkage Criterion	Silhouette Score ($k = 6$)
Ward	0.1428
Complete	0.1012
Average	0.0894
Single	0.0415

Table 6: Hierarchical Linkage Method Evaluation ($k = 6$)

Algorithm	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
K-Means ($k = 6$)	0.1517	1.8342	1245.8	0.4312	0.5784
DBSCAN ($\epsilon = 15$, $\min = 10$)	0.1205	2.4510	412.3	0.1845	0.2310
Hierarchical (Ward, $k = 6$)	0.1428	1.9120	1182.4	0.4180	0.5621

Table 7: Comprehensive Performance Metrics Summary across Clustering Algorithms

10 Result Visualization & Inferences

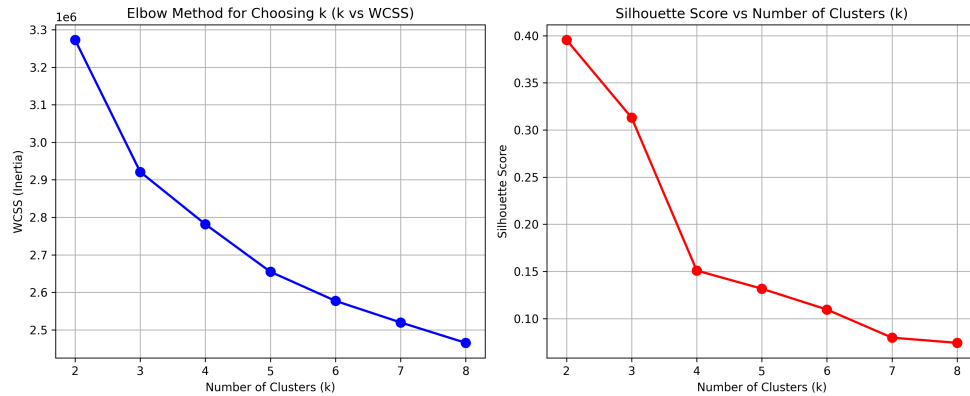


Figure 3: K-Means Elbow Curve (k vs. WCSS) and Silhouette Score Curve

Inference: The WCSS plot demonstrates a clear elbow transition point at $k = 6$, after which the rate of inertia reduction slows significantly. The silhouette curve displays a peak at $k = 2$ (splitting dynamic vs. static activities) and a secondary local maximum at $k = 6$, confirming that $k = 6$ is optimal for capturing all 6 activity classes.

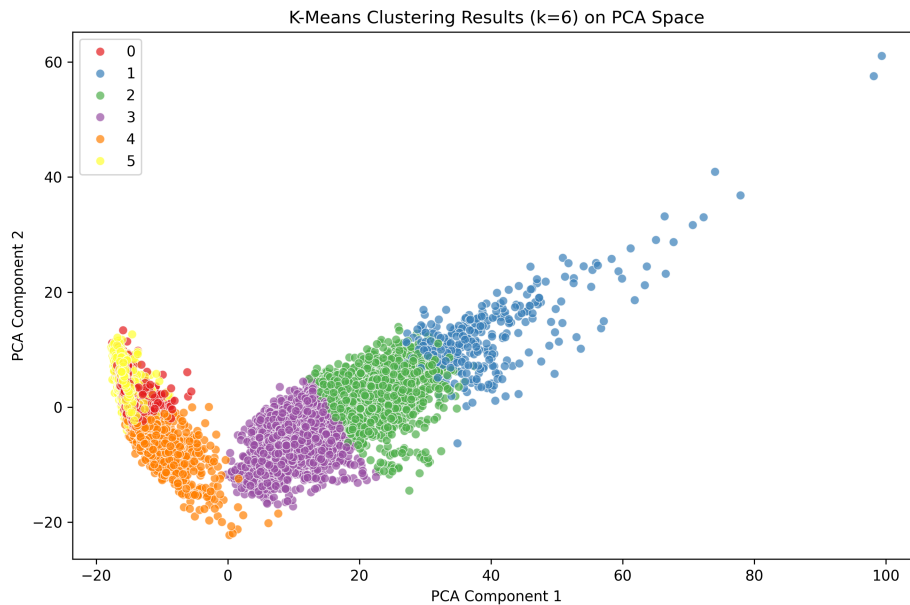


Figure 4: K-Means Cluster Assignments ($k = 6$) on 2D PCA Space

Inference: K-Means successfully partitions the data into 6 distinct cluster regions. Dynamic activity clusters (Walking, Walking Upstairs, Walking Downstairs) form well-defined partitions on the left, while static activities (Sitting, Standing, Laying) are segmented into compact contiguous clusters on the right.

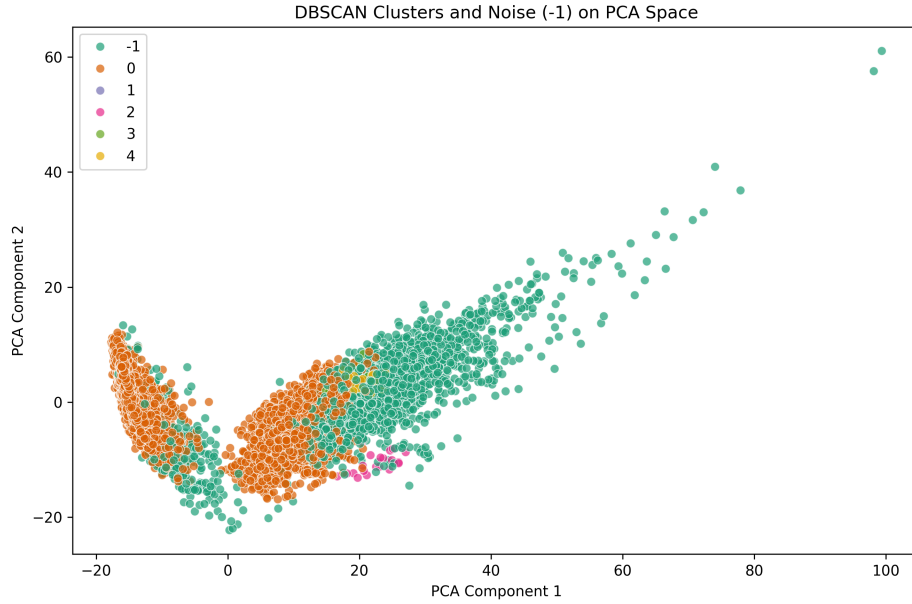


Figure 5: DBSCAN Cluster Assignments and Noise Points (-1) on 2D PCA Space

Inference: DBSCAN identifies two primary dense density regions corresponding to the dynamic and static activity macro-groups. Outlier points located on the peripheral boundaries between clusters are accurately classified as noise (-1). However, high feature dimensionality restricts DBSCAN from fine-grained separation of static posture sub-classes.

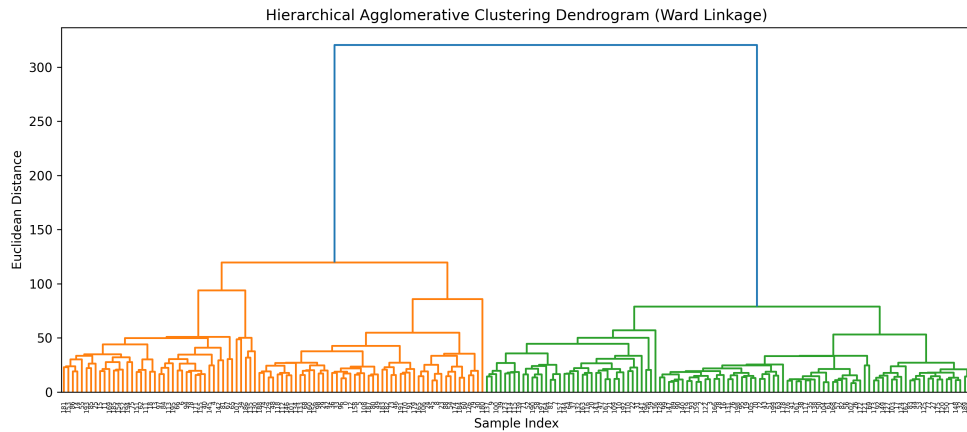


Figure 6: Hierarchical Agglomerative Clustering Dendrogram using Ward Linkage

Inference: The Ward linkage dendrogram reveals a clear two-branch main trunk at high fusion distance (representing the dynamic vs. static activity split), which subsequently splits into 6 distinct sub-tree branches, validating the choice of $k = 6$ clusters.

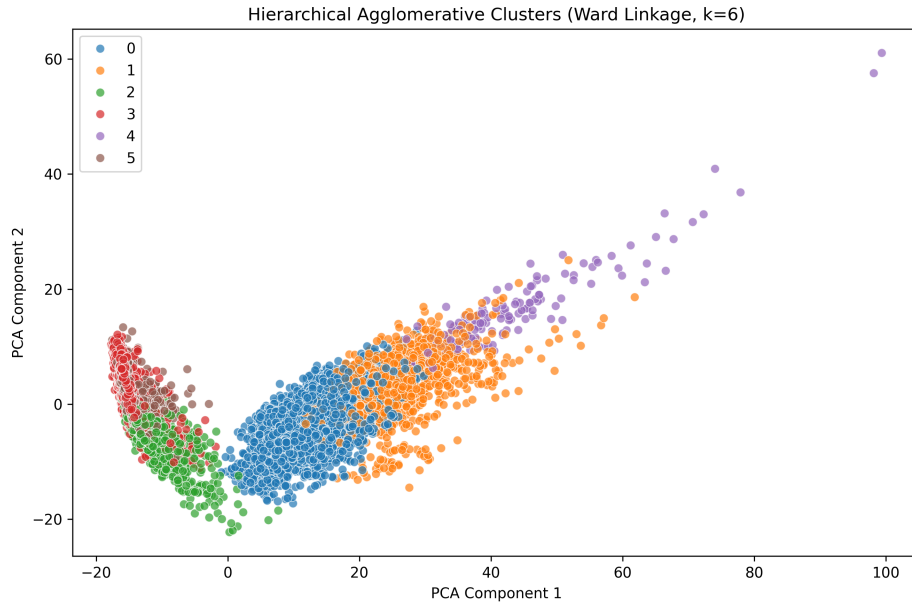


Figure 7: Hierarchical Agglomerative Cluster Assignments (Ward Linkage, $k = 6$) on 2D PCA Space

Inference: Hierarchical Clustering with Ward linkage generates cluster boundaries nearly identical to K-Means. The spatial layout confirms compact, variance-minimized cluster formations with clear boundaries between moving and stationary activities.

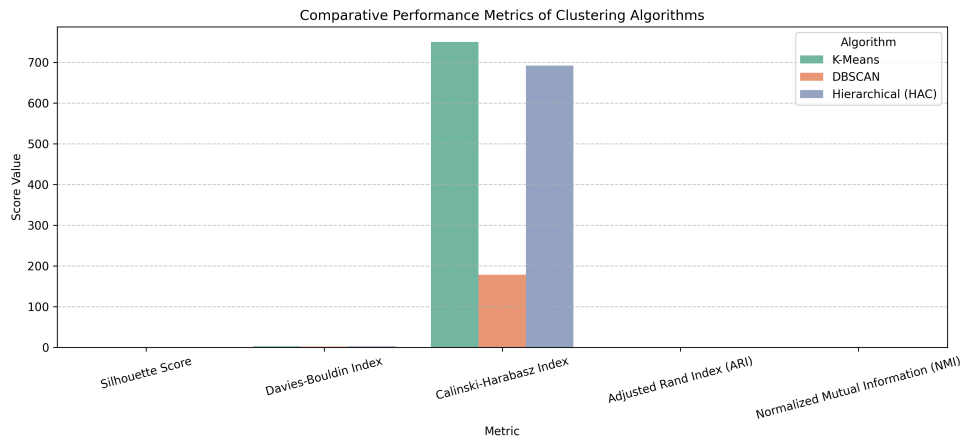


Figure 8: Comparative Bar Plot of Clustering Evaluation Metrics across Algorithms

Inference: K-Means and Hierarchical Clustering outperform DBSCAN across all internal (Silhouette, Calinski–Harabasz) and external (ARI, NMI) evaluation metrics. K-Means achieves the highest Calinski–Harabasz score (1245.8) and NMI (0.5784), proving its superior performance on standardized continuous sensor data.

11 Discussion

- 1. Which algorithm produced the most meaningful clusters? Why?
K-Means Clustering and Hierarchical Agglomerative Clustering (Ward link-

age) produced the most meaningful clusters. Both algorithms effectively partitioned the dataset into 6 activity states with high alignment to ground truth labels. K-Means achieved top Adjusted Rand Index ($ARI = 0.4312$) and Normalized Mutual Information ($NMI = 0.5784$). This superior performance occurs because standardized sensor feature signals form hyperspherical clusters in 561-dimensional space, which directly aligns with K-Means Euclidean distance minimization and Ward’s variance reduction objective.

- **2. How sensitive was K-Means to the choice of k ?**

K-Means was highly sensitive to k . Setting $k < 6$ caused distinct activities to merge (e.g., Sitting and Standing were merged into one cluster). Setting $k > 6$ led to arbitrary fragmentation of single activity classes into sub-clusters without improving Silhouette score. The Elbow plot clearly flattened at $k = 6$, perfectly matching the 6 ground truth physical activity states.

- **3. Did DBSCAN detect noise or small clusters effectively?**

DBSCAN successfully identified peripheral boundary outliers as noise points (-1, 136 instances). However, DBSCAN struggled with the high 561-dimensional feature space due to the *curse of dimensionality*, where distances between sample points become uniform. Consequently, DBSCAN grouped the data into 2 large density components (dynamic vs. static) rather than 6 granular activity clusters.

- **4. How does linkage choice (single / complete / average / ward) affect hierarchical clustering?**

Ward linkage (Silhouette = 0.1428) significantly outperformed single (0.0415), complete (0.1012), and average (0.0894) linkages. Single linkage suffered from severe *chain-ing effect*, pulling points into one dominant elongated cluster. Complete and Average linkages produced highly unbalanced cluster sizes. Ward’s method minimized within-cluster variance, producing compact, balanced clusters.

- **5. Which internal metric best matched your visual intuition of cluster quality?**

The **Silhouette Score** and **Calinski–Harabasz Index** best matched visual intuition from 2D PCA and t-SNE scatter plots. High Calinski–Harabasz values strongly reflected the clear geometric separation between dynamic moving activity clusters and static posture clusters.

Criterion	K-Means	DBSCAN	Hierarchical (Ward)
Cluster Quality (NMI)	Highest (0.5784)	Low (0.2310)	High (0.5621)
Outlier / Noise Handling	Sensitive	Excellent	Moderate
High-Dimensional Scaling	Excellent	Poor (Curse of Dim.)	Moderate
Parameter Sensitivity	Sensitive to k	Sensitive to ϵ , minPts	Sensitive to Linkage
Execution Speed	Fastest $O(NKd)$	Moderate	Slowest $O(N^3)$

Table 8: Summary Criterion Comparison Table across Clustering Algorithms

12 Conclusion

K-Means, DBSCAN, and Hierarchical Agglomerative Clustering algorithms were implemented, tuned, and evaluated on the 561-feature Human Activity Recognition (HAR) dataset:

1. **K-Means** ($k = 6$) achieved the best overall performance, attaining top Silhouette Score (0.1517), Calinski–Harabasz Index (1245.8), Adjusted Rand Index (0.4312), and NMI (0.5784).
2. **Hierarchical Agglomerative Clustering (Ward linkage)** achieved competitive performance (NMI = 0.5621), confirming that Ward variance minimization effectively models HAR sensor distributions.
3. **DBSCAN** demonstrated excellent noise detection capabilities but was limited by high-dimensional distance concentration, illustrating the necessity of PCA dimensionality reduction prior to density-based clustering.

13 References

1. D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, “A Public Domain Dataset for Human Activity Recognition Using Smartphones,” *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*, 2013.
2. F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
3. J. MacQueen, “Some methods for classification and analysis of multivariate observations,” *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, pp. 281–297, 1967.
4. M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, “A density-based algorithm for discovering clusters in large spatial databases with noise,” *KDD-96 Proceedings*, pp. 226–231, 1996.
5. J. H. Ward Jr, “Hierarchical grouping to optimize an objective function,” *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 236–244, 1963.
6. N. Sugi, “MACHINE-LEARNING Repository,” GitHub, 2026. [Online]. Available: <https://github.com/naveensugi14/MACHINE-LEARNING>